
FISHPLORE
EXPLORE • DISCOVER • LEARN
An Interactive Book About Malaysia's Fish Species
LET'S BEGIN
Play the animated book
fishplore.vercel.app

WELCOME ABOARD · DISCOVER MALAYSIA, ONE RIVER AT A TIME

Every river tells a story.

Malaysia is home to many incredible freshwater fish. Let's follow one river from its cool upper streams to the place where it meets the sea.

01 **Explore four river zones**
Hulu, Pertengahan, Hilir and Muara

02 **Meet fascinating fish**
Eight species profiles to open

03 **Play the river quiz**
Where do I belong?

FIELD PASS · OPTIONAL

Open the river map →

Type your explorer name in the box above.



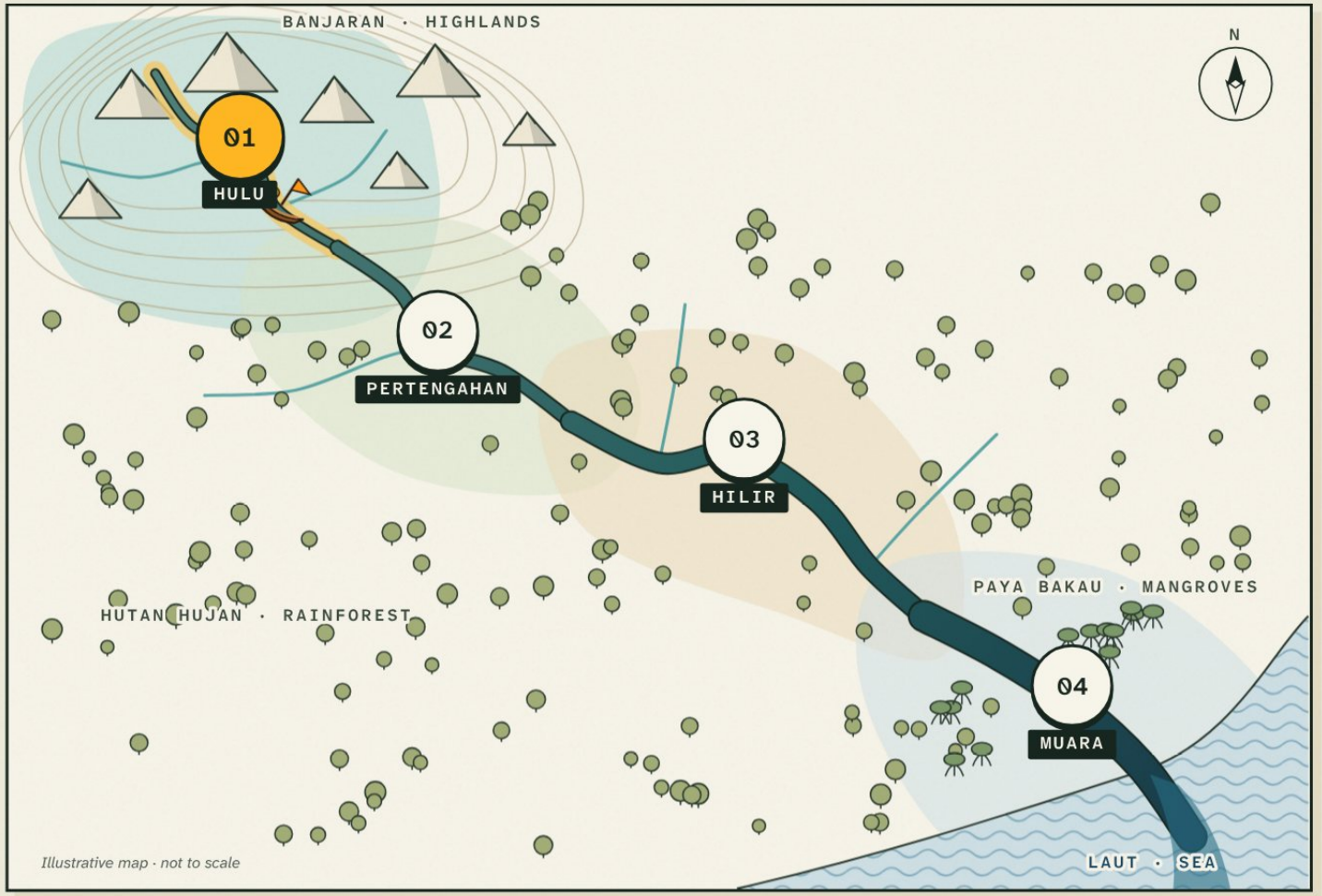
Kamus kecil LITTLE DICTIONARY

Sungai	river	Ikan	fish
Hulu	upper river	Pertengahan	middle
Hilir	downstream	Muara	river mouth

YOUR EXPLORATION MAP

Follow the river.

Start in the mountains. Choose a stop to jump into each river habitat.



STOP 01 · HULU

Upper River

hulu: the upper part of a river, near where it begins

Narrow, shaded streams tumble over boulders under the forest canopy.

Fast current | Rocky bed | Cooler streams

Fish to meet: Ikan Kelah, Ikan Tengas

Go to Stop 01 →

Track your journey

0 OF 4 STOPS VISITED

01 UPPER	02 MIDDLE	03 LOWER	04 MOUTH
Not yet	Not yet	Not yet	Not yet

STOP 01 · HULU

The Upper River.

Water begins its journey in forested highlands. Many upper reaches have faster currents and rocky beds.

Habitat notes

1 Fast current →

Water rushes downhill and splashes over rocks, mixing in plenty of oxygen.

2 Rocky bed →

Strong flow carries fine mud away, leaving boulders, cobbles and gravel behind.

3 Cooler streams →

Forest shade and higher ground help keep the water cool.

RIVER EXPLORER TIP

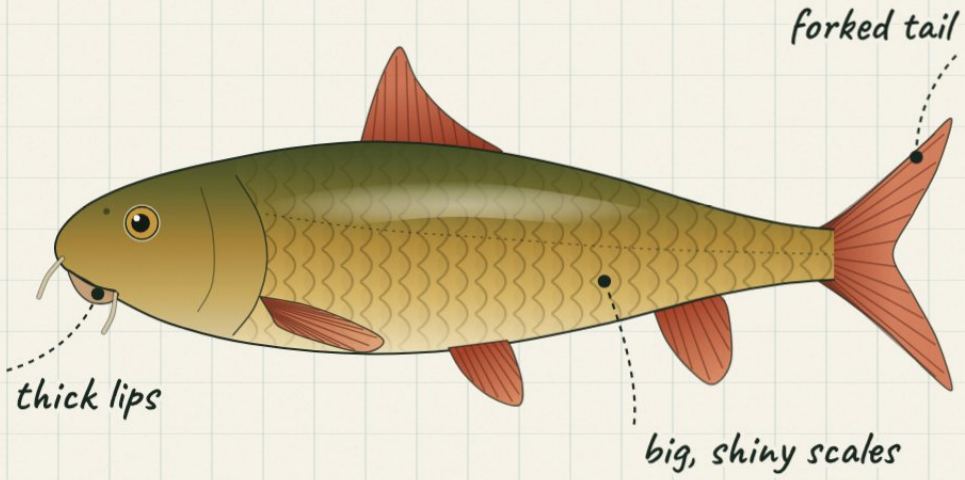
Look for rocks, flowing water and shaded riverbanks.

Meet the mountain swimmers →



Meet the mountain swimmers.

Two native fish to discover. Tap a fish to open its field profile.



FP-01 | NATIVE FRESHWATER FISH

Ikan Kelah

Tor tambra · Malaysian mahseer

A prized mahseer often associated with well-oxygenated, forested streams.

Open species profile →

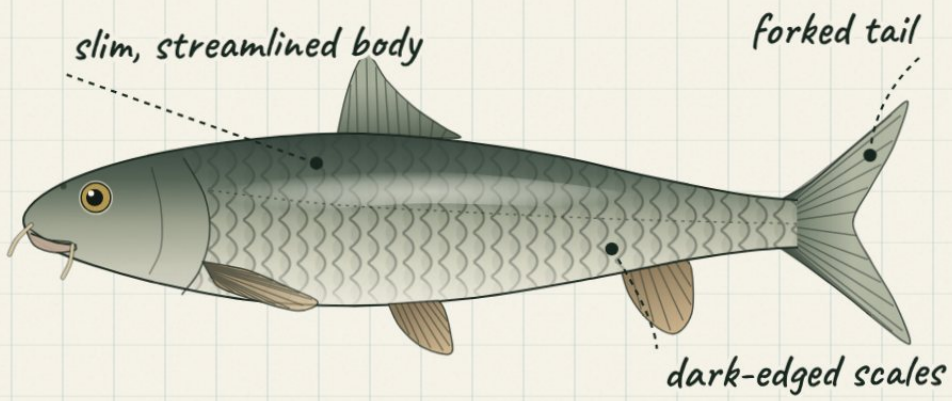
FP-02 | NATIVE FRESHWATER FISH

Ikan Tengas

Neolissochilus soroides

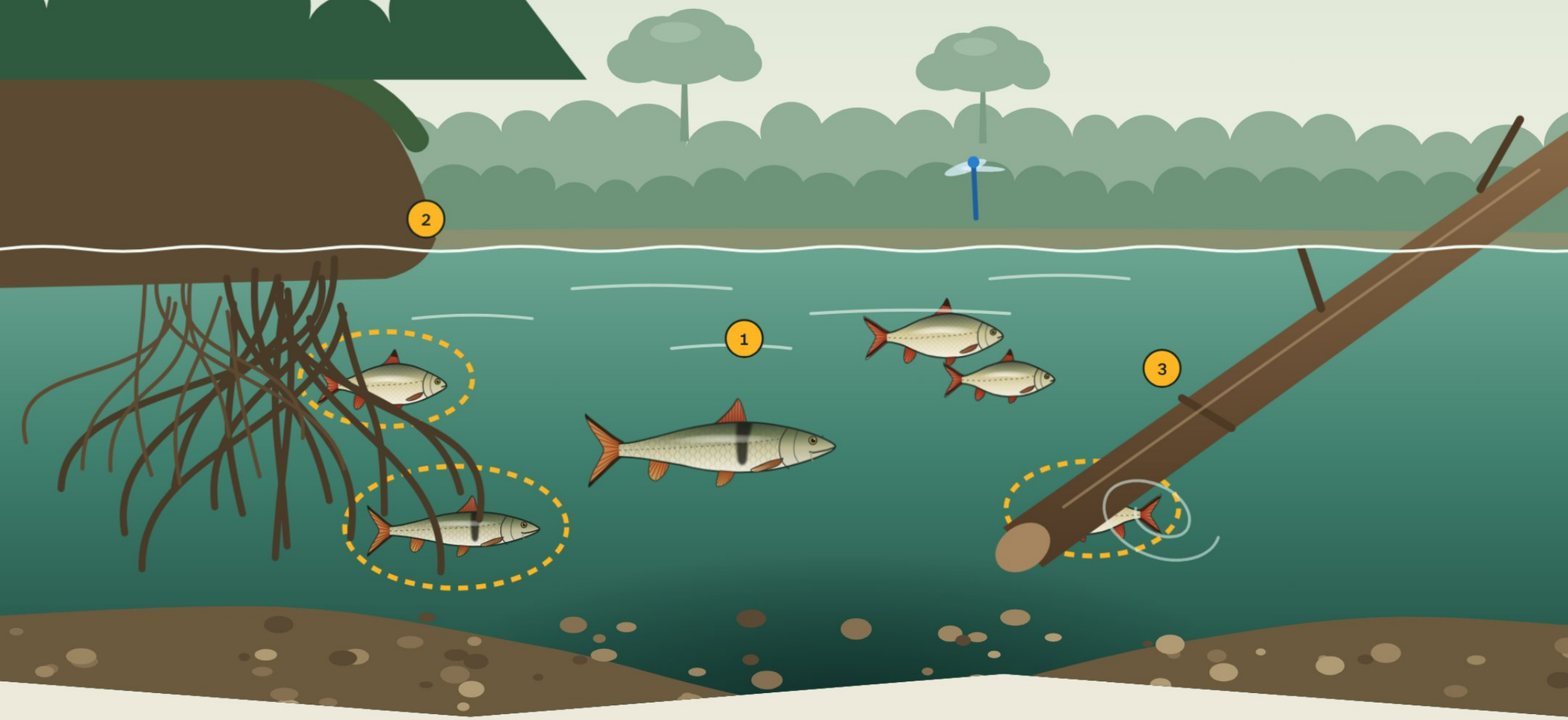
A native carp that can be found in clear, flowing rivers and streams.

Open species profile →



ⓘ Fish can use more than one stretch of a river. Kelah and tengas are often found in the upper river, but not only there.

Continue to Stop 02 →



STOP 02 · PERTENGAHAN

The Middle River.

The channel widens. Currents, pools, roots and fallen logs create different places for fish to feed and shelter.

Habitat notes

1 Mixed currents →

Fast runs and slow, deep pools sit side by side along the channel.

2 Riverbank roots →

Tree roots reach into the water and help hold the bank together.

3 More hiding places →

Fallen logs and tangled roots make shady shelters where fish rest and hide.

LOOK CLOSELY

Fish may hide near roots and logs.

Hidden fish found 3 of 3

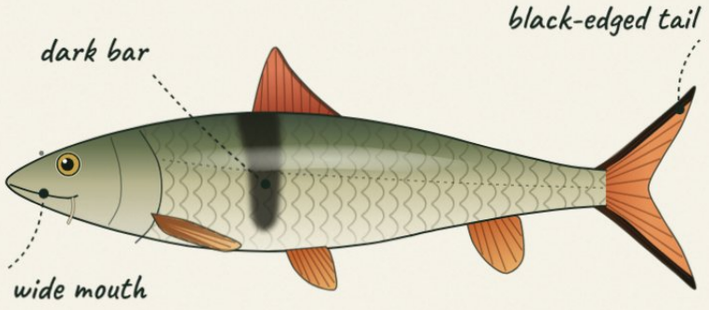
You found them all. Roots and logs make good shelter.

Discover the fish →

Fast fins & silver flashes.

Different body shapes and colours help us recognise river fish.

Handwritten notes point out what to look for on each fish.



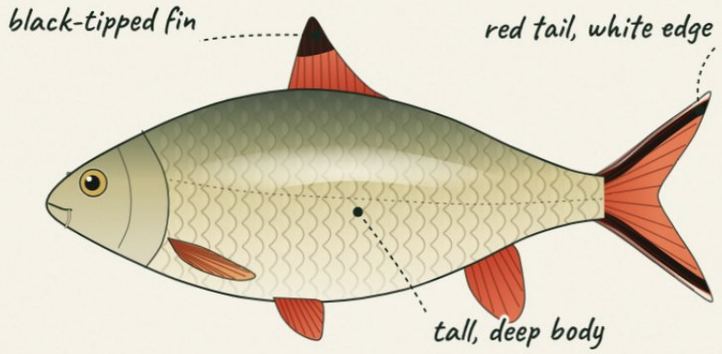
FP-03 | ACTIVE HUNTER

Ikan Sebarau

Hampala macrolepidota · Hampala barb

An active predator known for its dark body marking and swimming strength.

Open species profile →



FP-04 | SILVER FLASH

Lampam Sungai

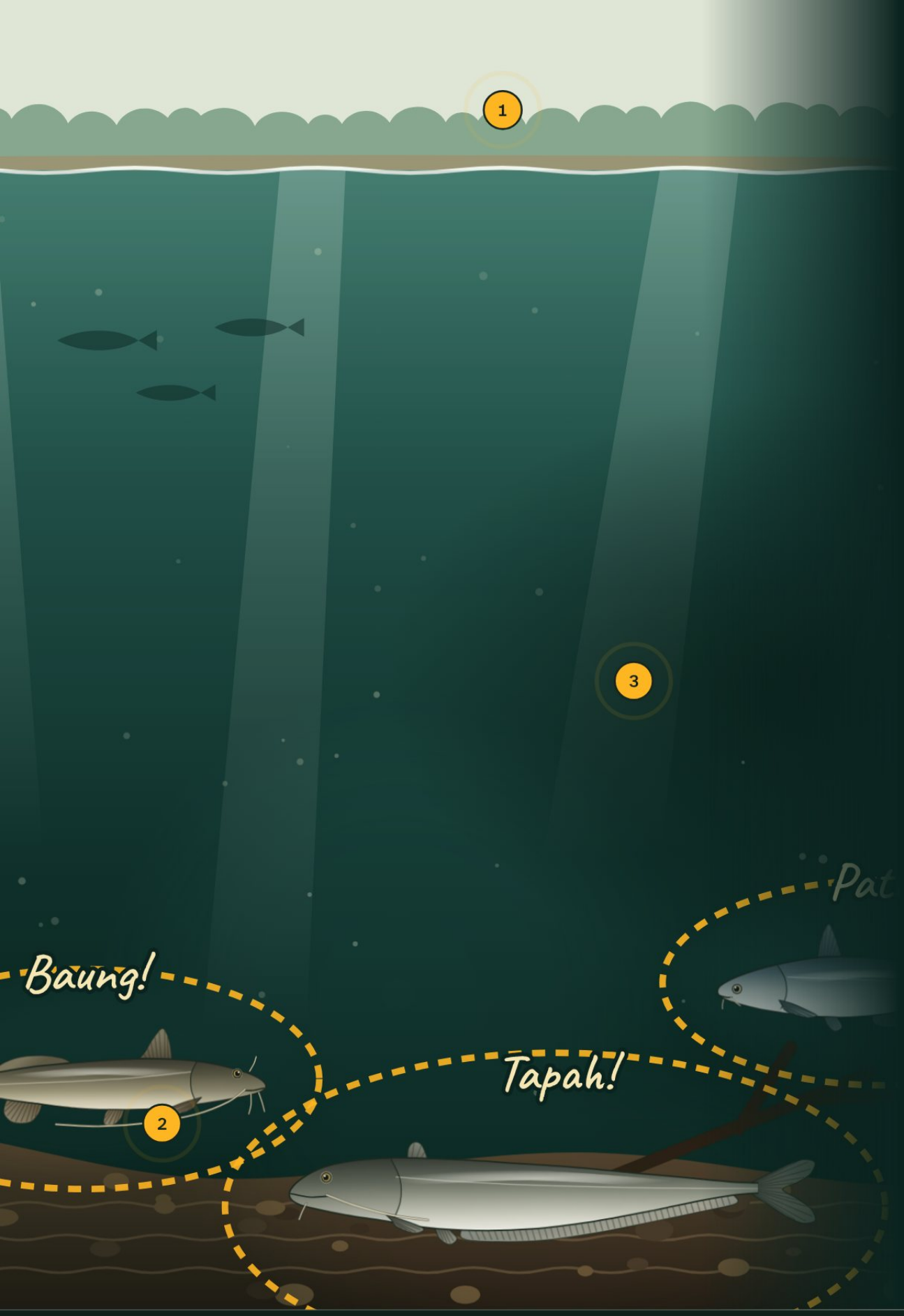
Barbonymus schwanefeldii · Tinfoil barb

A shiny, deep-bodied freshwater fish with colourful fins.

Open species profile →

ⓘ Want photos and records? Each profile links to Malaysian biodiversity information and fish databases.

Continue to Stop 03 →



STOP 03 · HILIR

The Lower River.

The river is often wider and deeper. Softer, muddy bottoms can support many fish that search for food near the bed.

Habitat notes

1 Wider waterway →

Streams have joined together, so the river carries much more water.

2 Soft riverbed →

Slower water lets fine silt and mud settle on the bottom.

3 Deeper pools →

Little light reaches the deepest water, where many fish feed near the bed.

EXPLORER MISSION

Find fish with whiskers!

Whiskered fish found 3 of 3

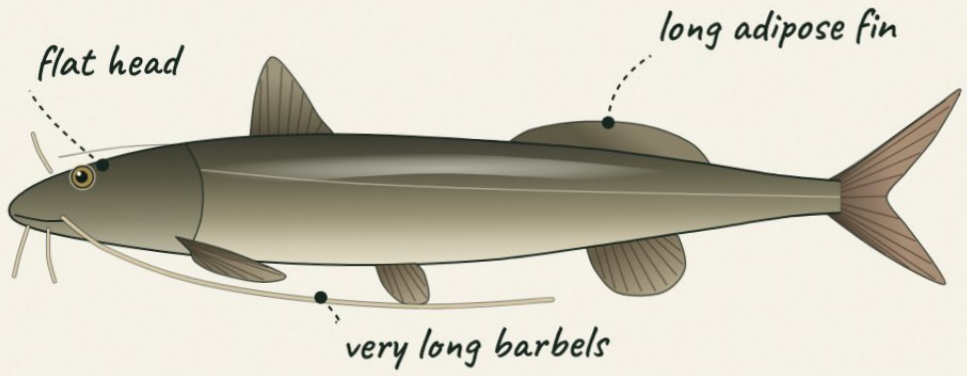
Mission complete! Baung, tapah and patin all feed near the riverbed.

Meet the deep dwellers →

Discover the river deep.

Catfish have whisker-like barbels that help them sense their surroundings.

01 SENSITIVE BARBELS



FP-05 | RIVER CATFISH

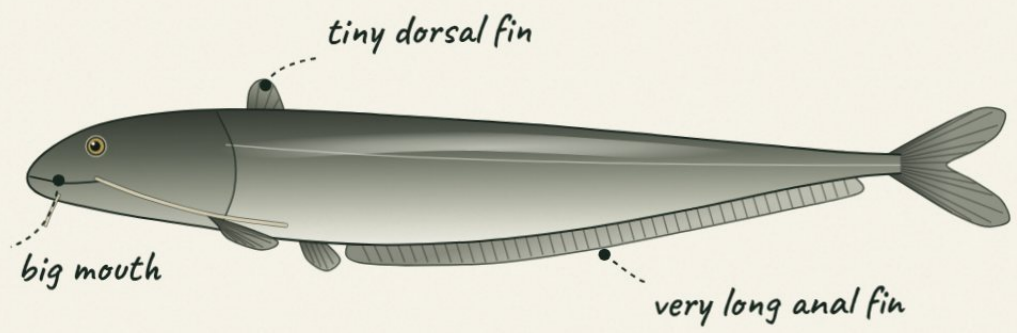
Ikan Baung

Hemibagrus nemurus

A river catfish with sensitive barbels.

Open species profile →

02 LONG, POWERFUL BODY



FP-06 | RIVER CATFISH

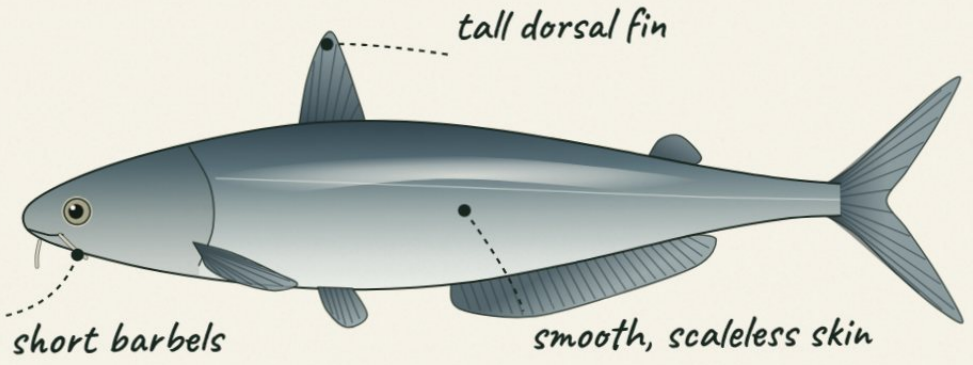
Ikan Tapah

Wallagonia leerii

A big catfish with a long, powerful body.

Open species profile →

03 SMOOTH-SKINNED RIVER FISH



FP-07 | RIVER CATFISH

Ikan Patin

Several species · Shark catfishes

Smooth-skinned river catfish.

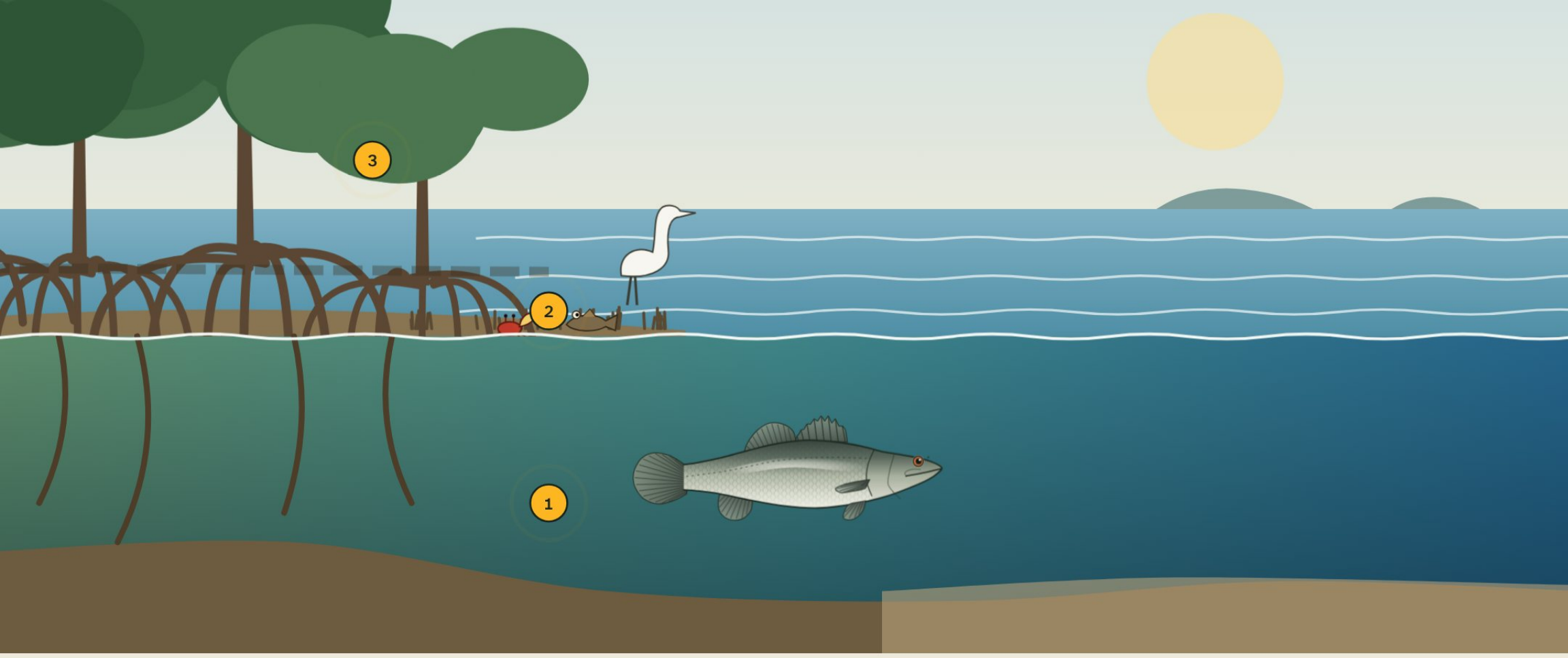
Open species profile →

One name, many fish

"Patin" is a common name for more than one catfish species, so there is no single scientific name for it. Patin buah (*Pangasius nasutus*) is a native patin, while the farmed *Pangasianodon hypophthalmus* was introduced to Malaysia. The drawing shows the general shape these shark catfishes share.

ⓘ Some carps have small barbels too. Catfish are named for their long, whisker-like ones.

Continue to Stop 04 →



STOP 04 · MUARA

Where River Meets Sea.

At the river mouth, fresh water mixes with seawater. Many estuaries contain brackish water, a unique transition zone.

Habitat notes

1 **Brackish water** →

River water and seawater mix, so the water is a little salty.

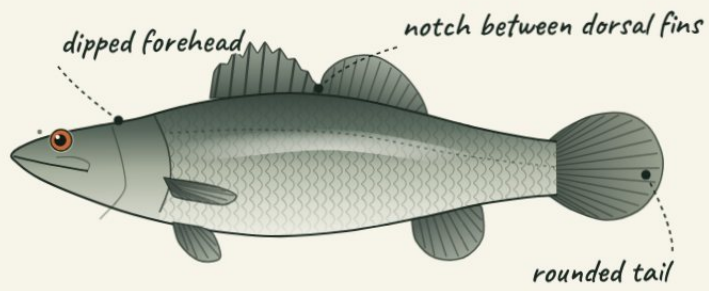
2 **Tides** →

Rising and falling tides push seawater in and out of the river mouth.

3 **Mangroves** →

Mangrove trees grow in salty mud, standing on tangled roots.

MEET: IKAN SIAKAP



FP-08 | FRESH, BRACKISH & SEA

Ikan Siakap

Lates calcarifer · Barramundi or Asian seabass

This fish can live in freshwater, brackish water and seawater.

Open species profile →

A simple salinity journey

From the river to the sea. Salinity means how salty the water is, measured in parts per thousand (ppt).



WATER

Freshwater

Rivers and lakes · **below 0.5 ppt**

SIAKAP HERE

Siakap do much of their growing in rivers and lakes connected to the sea.

Food and Agriculture Organization of the United Nations

WATER

Brackish water

Estuaries and mangrove creeks · **between 0.5 and 30 ppt**

SIAKAP HERE

Young siakap use brackish swamps and estuaries.

FishBase

WATER

Seawater

Coastal seas · **about 35 ppt**

SIAKAP HERE

Adult siakap travel to river mouths and coastal waters to spawn.

Food and Agriculture Organization of the United Nations

IMPORTANT

Not every fish found in an estuary is strictly a freshwater species.

Play the river quiz →

QUIZ WORKSHEET

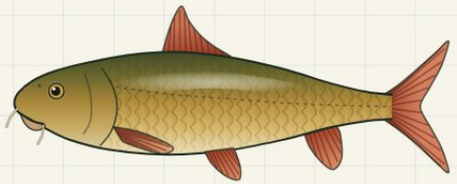
Where do I belong?

Tick the best-fit river zone for each fish. The answer key is on page 12.

QUESTION 1 | FP-01

Kelah

Tor tambra

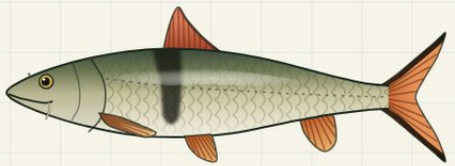


- ○ A — **Hulu** — Upper river
- ○ B — **Middle** — Pertengahan
- ○ C — **Hilir** — Lower river
- ○ D — **Muara** — River mouth

QUESTION 2 | FP-03

Sebarau

Hampala macrolepidota



- ○ A — **Hulu** — Upper river
- ○ B — **Middle** — Pertengahan
- ○ C — **Hilir** — Lower river
- ○ D — **Muara** — River mouth

QUESTION 3 | FP-06

Tapah

Wallagonia leerii

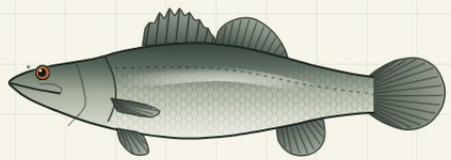


- ○ A — **Hulu** — Upper river
- ○ B — **Middle** — Pertengahan
- ○ C — **Hilir** — Lower river
- ○ D — **Muara** — River mouth

QUESTION 4 | FP-08

Siakap

Lates calcarifer



- ○ A — **Hulu** — Upper river
- ○ B — **Middle** — Pertengahan
- ○ C — **Hilir** — Lower river
- ○ D — **Muara** — River mouth



These are best-fit zones for our journey. Many fish use more than one stretch of a river!







RIVER EXPLORER

Adventure complete!

You followed the river, discovered native fish and explored a world worth protecting.

4/4 zones visited | 8/8 fish profiles opened | Take the river quiz

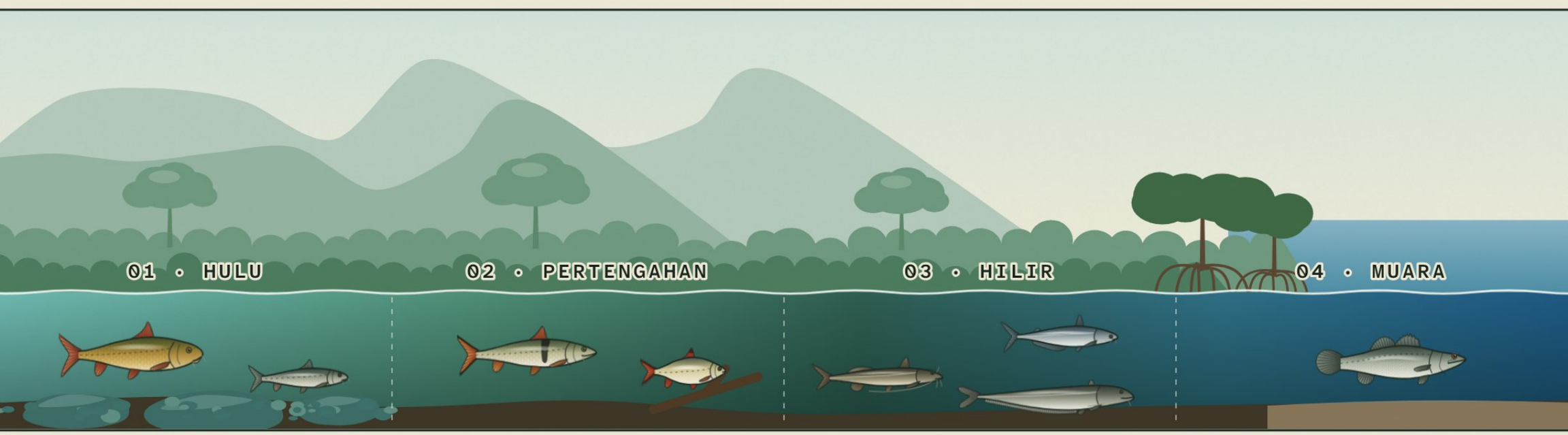
Protect our rivers

☐ I will keep waterways clean and respect river wildlife.

Tick the box to make your river pledge.

Explore again | Return to the river map

View fact-check sources



One river, from mountain stream to river mouth.

Quiz answer key

01 Kelah ..... A · Hulu

02 Sebarau ..... B · Middle

03 Tapah ..... C · Hilir

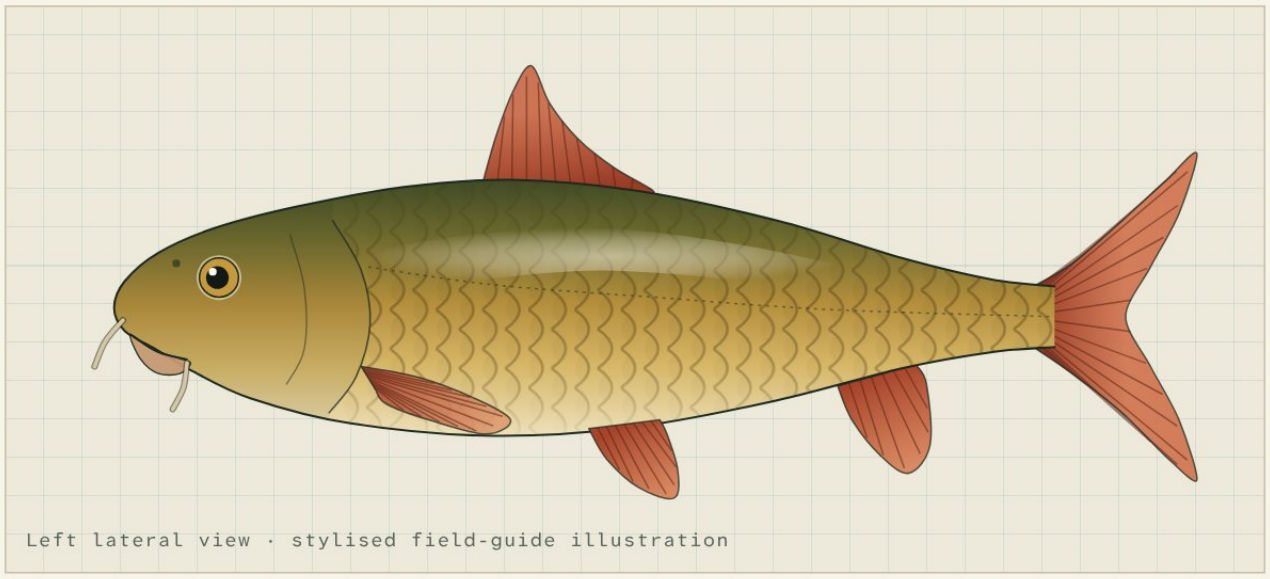
04 Siakap ..... D · Muara

ⓘ River zones are simplified; habitats overlap. Fish drawings are stylised, not identification guides.

SPECIMEN FP-01 | STOP 01 · HULU

Ikan Kelah

Tor tambra · Malaysian mahseer



Left lateral view · stylised field-guide illustration

A prized mahseer often associated with well-oxygenated, forested streams.

LOOK FOR

1. Large, shiny scales
2. Thick, fleshy lips
3. Two pairs of short barbels
4. Deeply forked tail

WHERE YOU MIGHT FIND IT

Clear, fast forest rivers. Adults rest in pools and runs over gravel and cobbles; young fish stay near rapids.

Best fit on this journey: **Upper River (Hulu)**

DID YOU KNOW?

> Kelah eat both plants and animals, including fruits that drop into the river from forest trees.
>
> FishBase

> Young kelah are found near rapids, while adults rest in pools and runs over gravel and cobbles.
>
> FishBase

> A big kelah can reach about 1 metre in length.
>
> FishBase

NAME CHECK

Malaysian fisheries sources often call kelah Tor tambroides. Recent taxonomic work treats that name as the same species as Tor tambra, though not every source has switched yet.

Eschmeyer's Catalog of Fishes: Tor tambroides

LEARN MORE

DEPARTMENT OF FISHERIES MALAYSIA
Spesies ikan tempatan (native fish species poster)

FISHBASE
Tor tambra species summary

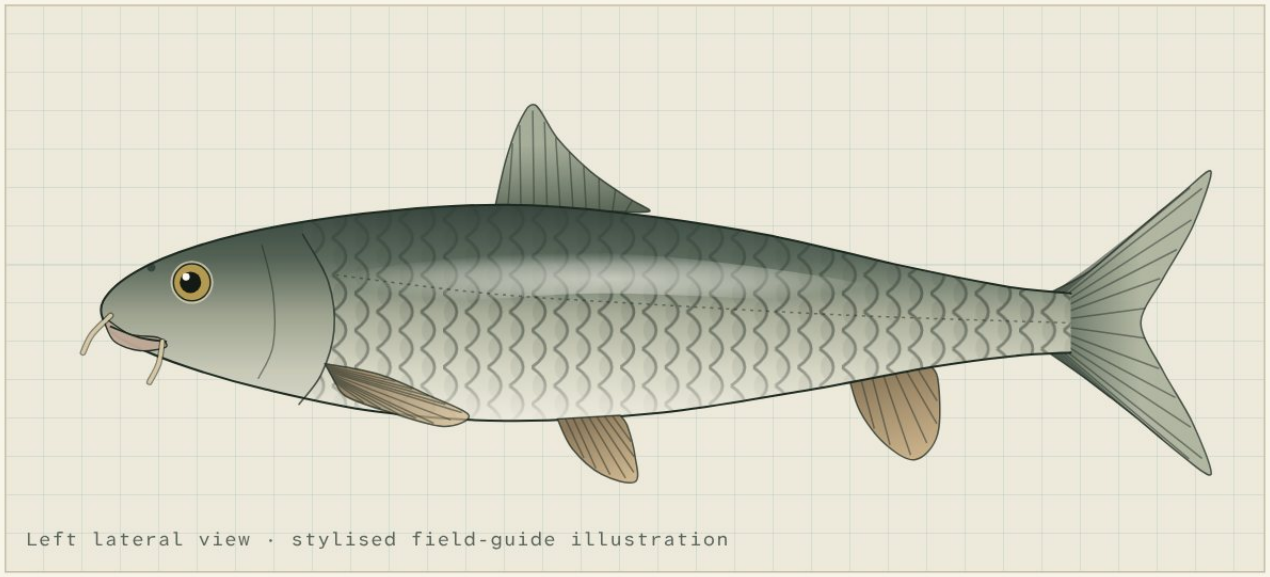
CALIFORNIA ACADEMY OF SCIENCES
Eschmeyer's Catalog of Fishes: Tor tambroides

Fish may use more than one stretch of a river. Drawings are stylised to show key features; they are not identification guides.

SPECIMEN FP-02 | STOP 01 · HULU

Ikan Tengas

Neolissochilus soroides



Left lateral view · stylised field-guide illustration

A native carp that can be found in clear, flowing rivers and streams.

LOOK FOR

1. Slimmer, more streamlined body than kelah
2. Large scales with darker edges
3. No dark stripe along the side
4. Forked tail

WHERE YOU MIGHT FIND IT

Pools in clear forest streams and rivers.

Best fit on this journey: **Upper River (Hulu)**

DID YOU KNOW?

> Tengas live in the pools of clear forest streams and rivers.
>
> FishBase

> Tengas can grow to about 45 cm long.
>
> FishBase

LEARN MORE

DEPARTMENT OF FISHERIES MALAYSIA
Spesies ikan tempatan (native fish species poster)

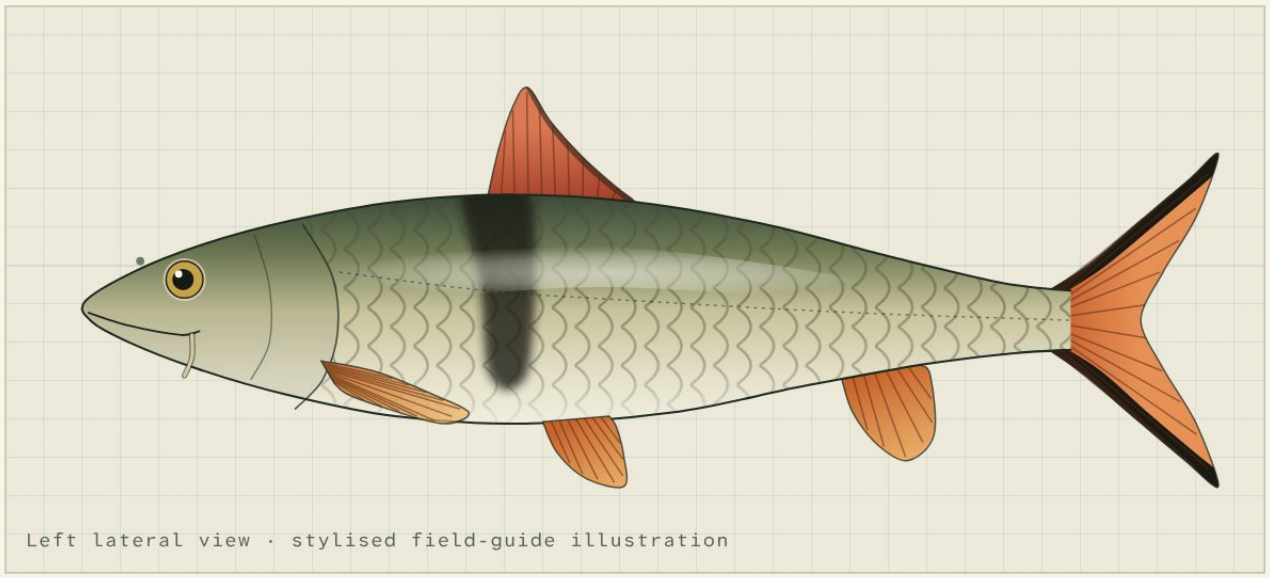
FISHBASE
Neolissochilus soroides species summary

Fish may use more than one stretch of a river. Drawings are stylised to show key features; they are not identification guides.

SPECIMEN FP-03 | STOP 02 · PERTENGAHAN

Ikan Sebarau

Hampala macrolepidota · Hampala barb



Left lateral view · stylised field-guide illustration

An active predator known for its dark body marking and swimming strength.

LOOK FOR

1. A black bar between the dorsal and pelvic fins
2. Orange-red tail with a black stripe along each lobe
3. Large head and wide mouth
4. Long, flattened-sided body

WHERE YOU MIGHT FIND IT

Found in most river waters except small creeks, torrents and shallow swamps.

Best fit on this journey: **Middle River (Pertengahan)**

DID YOU KNOW?

> Sebarau are hunters. Adults studied in a lake at Zoo Negara, Malaysia, ate mostly other fish.
>
> FishBase

> They live in most kinds of river water, but not in tiny creeks, torrents or shallow swamps.
>
> FishBase

> Sebarau is known for its dark body marking and its swimming strength.
>
> From the original book

LEARN MORE

MYBIS
Malaysia Biodiversity Information System (MyBIS)

FISHBASE
Hampala macrolepidota species summary

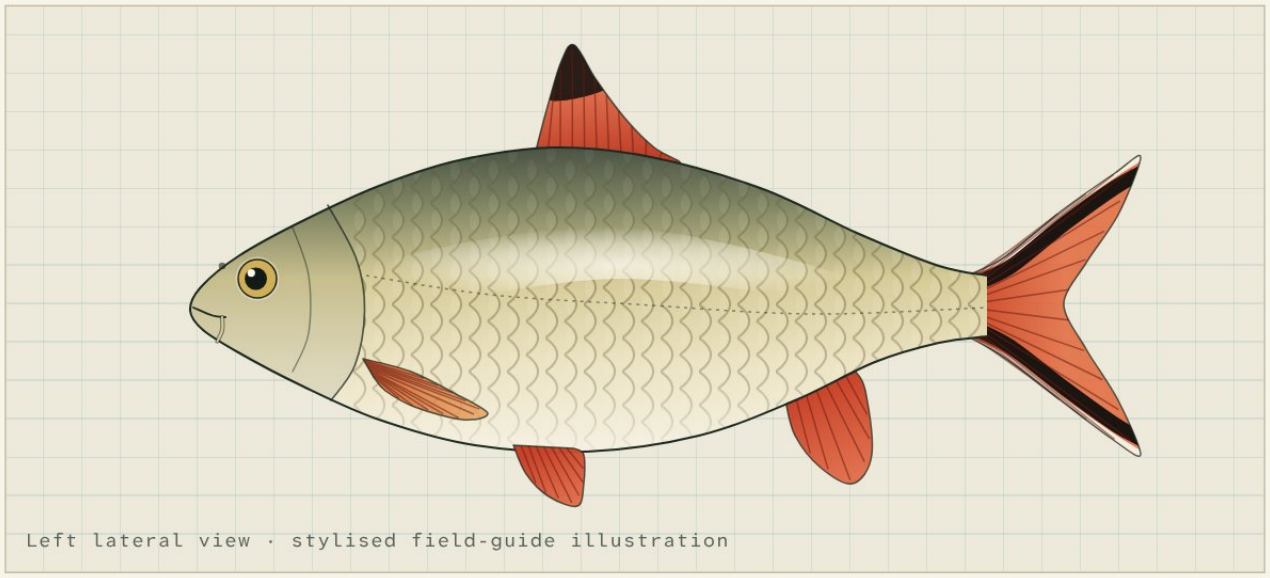
ECOLOGY ASIA
Sebarau (Hampala macrolepidota)

Fish may use more than one stretch of a river. Drawings are stylised to show key features; they are not identification guides.

SPECIMEN FP-04 | STOP 02 · PERTENGAHAN

Lampam Sungai

Barbonymus schwanefeldii · Tinfoil barb



Left lateral view · stylised field-guide illustration

A shiny, deep-bodied freshwater fish with colourful fins.

LOOK FOR

1. Tall, deep, silvery body
2. Red dorsal fin with a black tip
3. Red tail edged white with a black stripe inside
4. Small head and mouth

WHERE YOU MIGHT FIND IT

Rivers, streams and canals. Adults swim into flooded fields.

Best fit on this journey: **Middle River (Pertengahan)**

DID YOU KNOW?

> Lampam sungai mostly eat water plants and algae.
>
> FishBase

> They live in rivers, streams and canals, and adults move into flooded fields.
>
> FishBase

> Its red tail has a white edge and a black stripe just inside it on each lobe.
>
> FishBase

LEARN MORE

MYBIS
Malaysia Biodiversity Information System (MyBIS)

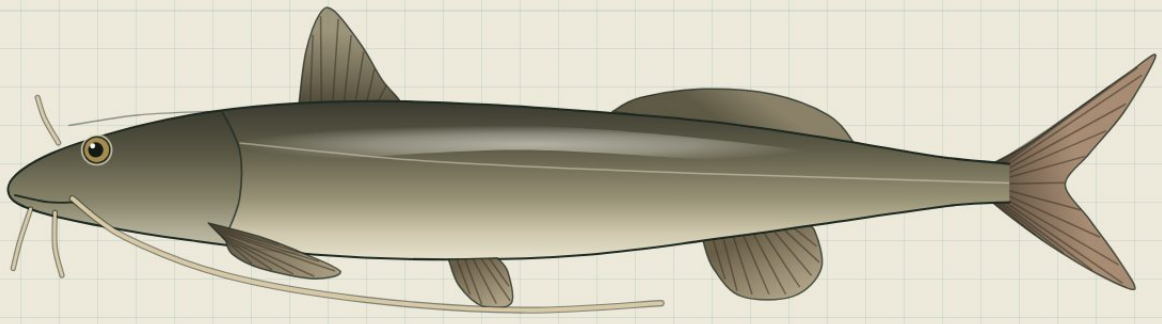
FISHBASE
Barbonymus schwanefeldii species summary

Fish may use more than one stretch of a river. Drawings are stylised to show key features; they are not identification guides.

SPECIMEN FP-05 | STOP 03 · HILIR

Ikan Baung

Hemibagrus nemurus



Left lateral view · stylised field-guide illustration

A river catfish with sensitive barbels.

LOOK FOR

1. Very long barbels beside the mouth
2. Flattened head
3. A long, fleshy adipose fin near the tail
4. Smooth skin without scales

WHERE YOU MIGHT FIND IT

Rivers with soft beds, where it searches for food close to the bottom.

Best fit on this journey: **Lower River (Hilir)**

DID YOU KNOW?

> Catfish like baung have whisker-like barbels that help them sense their surroundings near the riverbed.
>
> From the original book

> Baung have a long adipose fin, a soft fin with no rays, between the dorsal fin and the tail.
>
> Raffles Bulletin of Zoology 61: 205-291

NAME CHECK

Scientists revised this group in 2013 and limited the name Hemibagrus nemurus to Java. Baung in Peninsular Malaysia are often identified as Hemibagrus capitulum, so check the name used in newer guides.

Revision of the Asian catfish genus Hemibagrus (Ng & Kottelat, 2013)

LEARN MORE

DEPARTMENT OF FISHERIES MALAYSIA
Spesies ikan tempatan (native fish species poster)

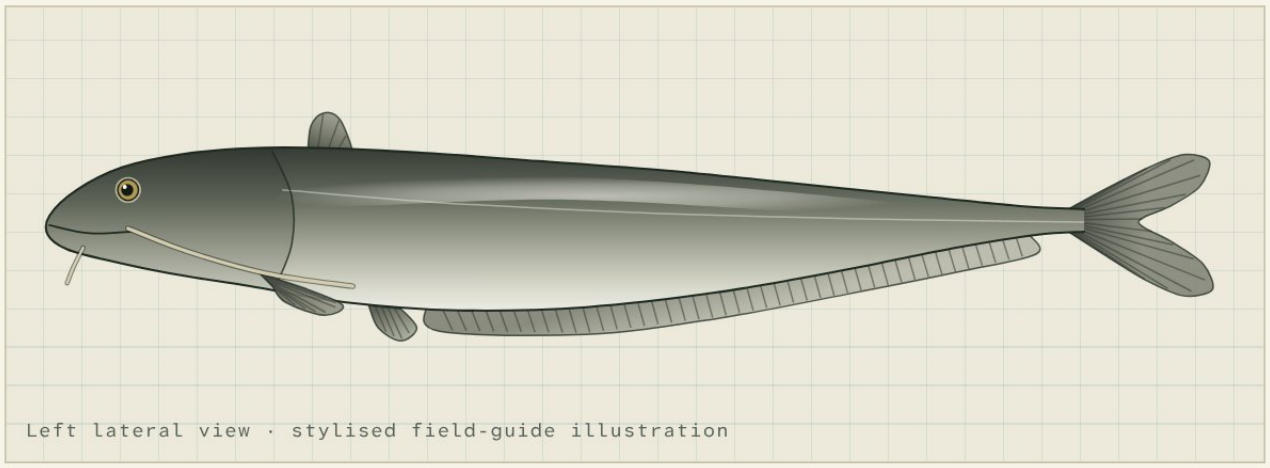
RAFFLES BULLETIN OF ZOOLOGY 61: 205-291
Revision of the Asian catfish genus Hemibagrus (Ng & Kottelat, 2013)

Fish may use more than one stretch of a river. Drawings are stylised to show key features; they are not identification guides.

SPECIMEN FP-06 | STOP 03 · HILIR

Ikan Tapah

Wallagonia leerii



Left lateral view · stylised field-guide illustration

A big catfish with a long, powerful body.

LOOK FOR

1. Very long anal fin under most of the body
2. Tiny dorsal fin
3. Large head and wide mouth
4. Two pairs of barbels

WHERE YOU MIGHT FIND IT

Big rivers. Adults move into flooded forest; young fish live where small streams join large rivers.

Best fit on this journey: **Lower River (Hilir)**

DID YOU KNOW?

> Tapah hunt fish and shrimp, mostly at night.
>
> FishBase

> Adults live in big rivers and move into flooded forest. Young tapah live where small streams meet big rivers.
>
> FishBase

> Tapah can grow longer than 1.5 metres.
>
> FishBase

NAME CHECK

Some references, including FishBase, still list tapah as Wallago leerii. Wallagonia leerii is the newer name for the same fish.

Wallago leerii (Wallagonia leerii) species summary

LEARN MORE

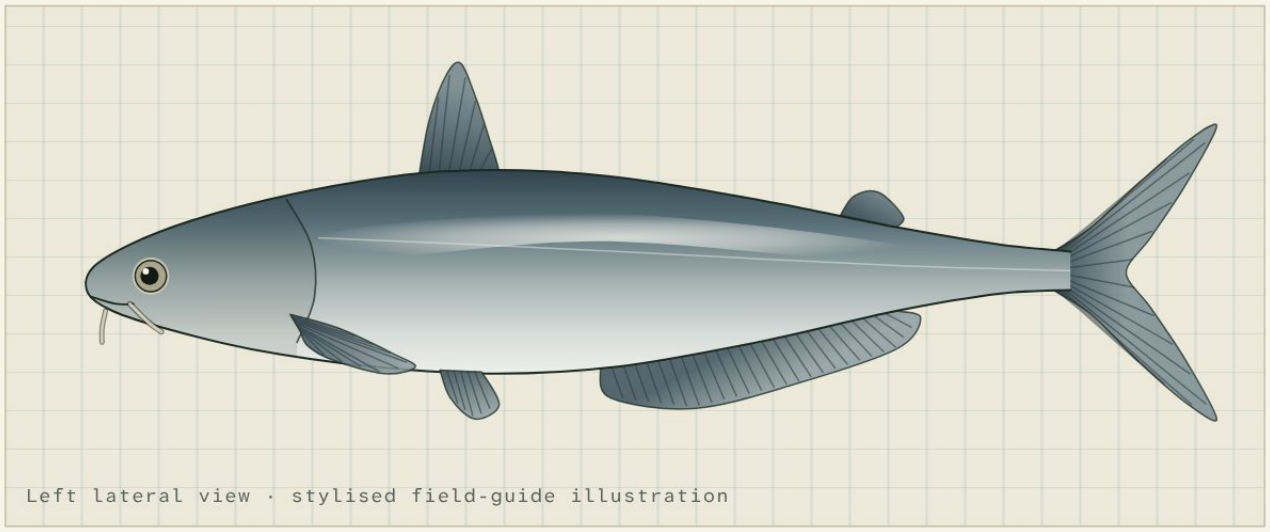
FISHBASE
Wallago leerii (Wallagonia leerii) species summary

Fish may use more than one stretch of a river. Drawings are stylised to show key features; they are not identification guides.

SPECIMEN FP-07 | STOP 03 · HILIR

Ikan Patin

Several species · Shark catfishes



Left lateral view · stylised field-guide illustration

Smooth-skinned river catfish.

LOOK FOR

1. Smooth, scaleless skin
2. Tall dorsal fin set far forward
3. Two pairs of short barbels
4. Small adipose fin and deeply forked tail

WHERE YOU MIGHT FIND IT

Large rivers. Exact habits depend on which patin species you mean.

Best fit on this journey: **Lower River (Hilir)**

DID YOU KNOW?

> "Patin" is a common name for more than one catfish species, so it is not one single fish.
>
> From the original book

> Patin buah (Pangasius nasutus) is a native patin of Malaysian rivers such as the Pahang River.
>
> Sains Malaysiana 43(11), 2014

> A patin often raised in fish farms, Pangasianodon hypophthalmus, was brought into Malaysia from Thailand. It is not native.
>
> Directory of Open Access Journals

LEARN MORE

SAINS MALAYSIANA 43(11), 2014
Study of patin buah, Pangasius nasutus, in the Pahang River

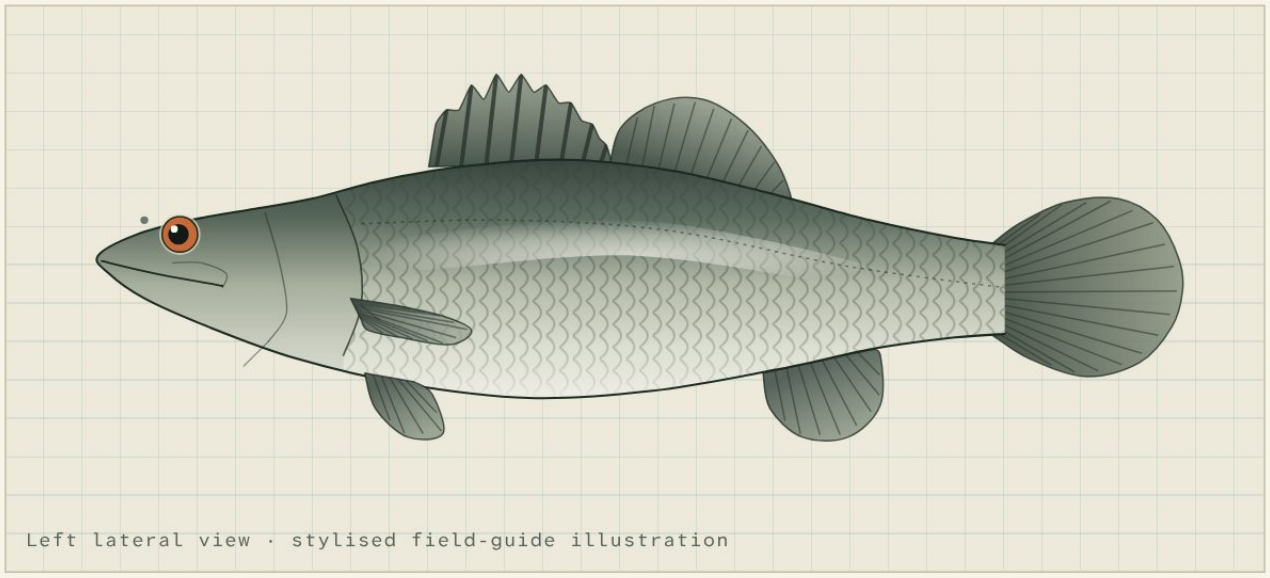
DIRECTORY OF OPEN ACCESS JOURNALS
Pangasianodon hypophthalmus in Malaysia

Fish may use more than one stretch of a river. Drawings are stylised to show key features; they are not identification guides.

SPECIMEN FP-08 | STOP 04 · MUARA

Ikan Siakap

Lates calcarifer · Barramundi or Asian seabass



Left lateral view · stylised field-guide illustration

This fish can live in freshwater, brackish water and seawater.

LOOK FOR

1. Pointed head with a dipped forehead
2. Large mouth reaching past the eye
3. Spiny and soft dorsal fins split by a deep notch
4. Rounded tail

WHERE YOU MIGHT FIND IT

Rivers, estuaries and coastal seas. It moves between them during its life.

Best fit on this journey: **River Mouth (Muara)**

DID YOU KNOW?

> Siakap grow up in rivers, then adults travel to river mouths and coastal waters to spawn.
>
> Food and Agriculture Organization of the United Nations

> Young siakap use brackish swamps and estuaries. Older juveniles move upstream into rivers.
>
> FishBase

> Many siakap start life as males and later become females.
>
> FishBase

LEARN MORE

FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS (FAO)
Biology of the seabass (barramundi), Lates calcarifer

FAO
Seabass (Lates calcarifer) hatchery and culture manual

FISHBASE
Lates calcarifer species summary

Fish may use more than one stretch of a river. Drawings are stylised to show key features; they are not identification guides.

Fact-check sources

Where the facts in this book come from. Each link opens in a new tab.

FROM THE ORIGINAL FISHPLORE BOOK

DEPARTMENT OF FISHERIES MALAYSIA
Spesies ikan tempatan (native fish species poster)
Used for: Native fish information

MYBIS
Malaysia Biodiversity Information System (MyBIS)
Used for: Malaysian biodiversity records and photographs

FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS (FAO)
Biology of the seabass (barramundi), Lates calcarifer
Used for: Barramundi life cycle and salinity

FURTHER REFERENCES FOR THIS EDITION

FAO
Seabass (Lates calcarifer) hatchery and culture manual
Used for: Adults moving to salty river mouths to spawn

FISHBASE
Tor tambra species summary
Used for: Kelah size and habitat

FISHBASE
Tor tambroides species summary
Used for: Kelah habitat and diet (account filed under Tor tambroides)

CALIFORNIA ACADEMY OF SCIENCES
Eschmeyer's Catalog of Fishes: Tor tambroides
Used for: Tor tambroides treated as a synonym of Tor tambra

FISHBASE
Neolissochilus soroides species summary
Used for: Tengas habitat and size

FISHBASE
Hampala macrolepidota species summary
Used for: Sebarau habitat and diet

ECOLOGY ASIA
Sebarau (Hampala macrolepidota)
Used for: Sebarau body shape and markings

FISHBASE
Barbonymus schwanefeldii species summary
Used for: Lampam sungai habitat, diet and fin colours

RAFFLES BULLETIN OF ZOOLOGY 61: 205-291
Revision of the Asian catfish genus Hemibagrus (Ng & Kottelat, 2013)
Used for: Naming of baung (Hemibagrus) species

FISHBASE
Wallago leerii (Wallagonia leerii) species summary
Used for: Tapah habitat, diet and size

SAINS MALAYSIANA 43(11), 2014
Study of patin buah, Pangasius nasutus, in the Pahang River
Used for: Patin buah as a native Malaysian catfish

DIRECTORY OF OPEN ACCESS JOURNALS
Pangasianodon hypophthalmus in Malaysia
Used for: Farmed patin introduced from Thailand

FISHBASE
Lates calcarifer species summary
Used for: Siakap habitat, life cycle and size

NOTES ON NAMES

Kelah	Shown as Tor tambra, as in the original book. Malaysian sources often use Tor tambroides, which recent taxonomy treats as the same species.
Baung	Shown as Hemibagrus nemurus, as in the original book. A 2013 revision limits that name to Java; Peninsular Malaysian baung are often identified as Hemibagrus capitulum.
Tapah	Wallagonia leerii. Some references, including FishBase, still use the older name Wallago leerii.
Patin	A common name for several catfish, not one species. Patin buah (Pangasius nasutus) is native; the farmed Pangasianodon hypophthalmus was introduced.
Lampam sungai	Spelled Barbonymus schwanefeldii here, following FishBase. Some books spell it schwanenfeldii.

SALINITY NUMBERS

Salinity is measured in parts per thousand (ppt). River water is usually below 0.5 ppt and open seawater is around 35 ppt. Brackish water falls in between. The original book labelled fresh water “0 ppt”; this edition uses “below 0.5 ppt”, the usual scientific boundary.

ⓘ River zones are simplified and habitats overlap. Fish drawings are stylised, not identification guides.